

- LO1:** Identify the characteristics and components of academic research
- LO2:** Analyse research problems in various IT fields
- LO3:** Evaluate research activities for ethical/unethical conduct
- LO4:** Identifying different sources of information
- LO5:** Evaluate the quality/reliability of different sources
- LO6:** Apply referencing styles in the acknowledgement of sources
- LO7:** Evaluate different research strategies
- LO8:** Analysing a conference call
- LO9:** Evaluate different data collection and analysis methods for specific research scenarios
- LO10:** Understanding how the abstract is constructed

QUESTION 1: MULTIPLE-CHOICE [60]

1. Answer: b) To gather and analyze data systematically	[2]
2. Answer: b) Aims to solve practical problems	[2]
3. Answer: b) A tentative statement to be tested	[2]
4. Answer: c) Non-numerical data	[2]
5. Answer: c) Identifying a research problem	[2]
6. Answer: d) Discussion	[2]
7. Answer: b) Can withdraw anytime without penalty	[2]
8. Answer: b) Using someone else's work without credit	[2]
9. Answer: a) Participants' identities are protected	[2]
10. Answer: b) Ensuring ethical compliance in studies	[2]
11. Answer: d) Technology and engineering journals	[2]
12. Answer: b) Only lists cited sources	[2]
13. Answer: b) A textbook summarizing a study	[2]
14. Answer: b) Reusing previous work without disclosure	[2]
15. Answer: b) Narrows results by requiring all terms	[2]
16. Answer: c) IEEE Xplore	[2]
17. Answer: b) Evaluated by experts before publication	[2]
18. Answer: b) Economic losses	[2]
19. Answer: b) Predicting the severity of traffic accidents	[2]
20. Answer: b) To prevent bias toward the majority class and improve minority class prediction (SMOTE was used to address imbalance, which is critical for unbiased severity classification.)	[2]
21. Answer: c) Other features (e.g., speed, road type) dominate severity prediction (Low feature importance suggests that other predictors are more influential, not necessarily that weather is irrelevant.)	[2]
22. Answer: b) Recall (to capture as many true fatalities as possible) (In safety-critical applications, missing severe cases is costlier than false positives.)	[2]
23. Answer: b) The performance gap between ensemble methods is small (A 1.5% improvement suggests RF is still competitive, not obsolete.)	[2]
24. Answer: d)	[2]

25. Answer: d)	[2]
26. Answer: a)	[2]
27. Answer: a)	[2]
28. Answer: b)	[2]
29. Answer: a)	[2]
30. Answer: b)	[2]

QUESTION 2: TRUE/FALSE [40]

31. Answer: False: B	[2]
32. Answer: True: A	[2]
33. Answer: True: A	[2]
34. Answer: False: B	[2]
35. Answer: True: A	[2]
36. Answer: True: A	[2]
37. Answer: False: B	[2]
38. Answer: False: B	[2]
39. Answer: False: B	[2]
40. Answer: False: B	[2]
41. Answer: False: B	[2]
42. Answer: True: A	[2]
43. Answer: False: B	[2]
44. Answer: True: A	[2]
45. Answer: False: B	[2]
46. Answer: False: B	[2]
47. Answer: False: B	[2]
48. Answer: False: B	[2]
49. Answer: True: A	[2]
50. Answer: False: B	[2]